local filesystem or a central database. A blockchain application stores the following on the blockchain node (on-chain):

- Transactions executed and confirmed
- Results of smart contract function execution
- State changes (changes in *storage* variable values)
- Logs of events emitted

These data are stored in designated data structures on a blockchain node and propagated to other stakeholder nodes as specified by a blockchain protocol.

> **DEFINITION** *On-chain data* is a set of information generated by transactions initiated by blockchain-based applications and the items used during blockchain materialization. Most of this data is stored in a block and its header.

Figure 6.1 compares a traditional application and a blockchain-based application. The traditional system on the left has data stored in a filesystem or database. On the right is a blockchain application, with its familiar `<Dapp>-app` and `<Dapp>-contract` parts. In general, Dapp contract-generated data is stored on-chain, and data created and used by the Dapp app is stored off-chain. The function calls from a Dapp invoke a smart contract, the Txs are generated, and related artifacts are recorded on the blockchain. In figure 6.1, follow the arrows from the `<Dapp>-app` to understand the on-chain data. In this chapter, you'll explore this relationship between the on-chain and off-chain data.